

Split Complex Lorentzian Dynamics

Part XXIV: Intrinsic Oriented Frames, Reference Independence, and the Lifted-Frame Frontier

From a basis-free frame carrier to a two-sheeted lift, with the unresolved internal torsor action isolated for Task XXV

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Abstract

Part XXIII completed the local reconstruction of the certified double cover: the internal core $\mathcal{L}$ projects onto the visible group G_{vis} with kernel $\{\pm 1\}$, admits explicit local sections and locally constant transition signs, and can be reconstructed exactly from those local data. Only after that reconstruction was the projection identified formally as a two-sheeted covering map [7, 1]. That left a sharper question. A group double cover is not yet a spin structure, and a covering group by itself does not say what geometric object is to be lifted. Task 24 therefore moves one level upward while deliberately avoiding a varying base manifold: it asks whether the already reconstructed three-dimensional metric carrier contains, intrinsically, the oriented frame object on which the visible group acts.

The answer is substantially positive. Task 24 defines the carrier $\mathcal{F}^+(E)$ of ordered positively oriented orthonormal triples using only the inherited spatial carrier E , its positive-definite form, and a minimal orientation datum reconstructed from the inherited spatial triple product. Neither G_{vis} nor a reference frame occurs in this definition. A coordinate triple is used only to prove nonemptiness. Every frame is proved to be a basis. The inherited visible action is then lifted componentwise to $\mathcal{F}^+(E)$, where it preserves orthonormality and orientation. The action is formally proved free and transitive; consequently, for any two frames F, G , there exists exactly one $g \in G_{\text{vis}}$ with $g \cdot F = G$. A reference frame F_0 is introduced only afterwards and yields a coordinate equivalence $G_{\text{vis}} \simeq \mathcal{F}^+(E)$. Changing the reference frame changes this coordinate presentation by exactly one right translation. Thus the reference frame coordinatizes the intrinsic frame carrier but does not define it.

The certified projection $p : \mathcal{L} \rightarrow G_{\text{vis}}$ is then used to construct a lifted-frame carrier $\tilde{\mathcal{F}}_{F_0}$ from compatible pairs (u, F) satisfying $p(u) \cdot F_0 = F$. Every ordinary frame has exactly two lifted representatives; two lifts of the same frame differ by a unique kernel sign; the kernel-sign action is free; and the quotient by that sign relation is equivalent to $\mathcal{F}^+(E)$. For two reference frames the corresponding lifted carriers are explicitly equivalent, and the equivalence preserves the ordinary frame and the kernel-sign action. The price of moving upstairs is visible exactly here: the change of reference downstairs is unique, whereas choosing an internal lift of that visible reference change introduces a residual two-valued ambiguity. Task 24 proves that two such comparison maps differ by a kernel sign.

The formal run is green and internally closed, but a post-run review exposes one important boundary in the interpretation of Package I. Task 24 correctly proves that the direct $\mathcal{L}$ -action on the *ordinary* frame carrier, defined through p , is transitive but not free: its stabilizer is exactly $K_{\pm} = \{\pm 1\}$. The audit then describes the bijection $\tilde{\mathcal{F}}_{F_0} \simeq \mathcal{L}$ as the strongest available comparison and does not formalize a $\mathcal{L}$ -action on $\tilde{\mathcal{F}}_{F_0}$ itself. These are distinct statements. Nothing in Task 24 proves that a lifted-frame action is non-free; rather, that action was never defined. At paper level there is a natural candidate,

$$a \cdot (u, F) = (au, p(a) \cdot F),$$

which appears algebraically to be free and transitive and to make the lifted-frame projection equivariant. This is a diagnostic consequence of the completed definitions, not a Task-24 Lean theorem. It must therefore be hardened in a new task rather than retroactively credited to Task 24.

Task 25 is consequently a focused hardening task, not a new branch. It should formalize the internal action on the lifted-frame carrier, prove freeness, transitivity and projection equivariance, identify the restriction to $K_{\pm}$ with the already proved vertical sign action, make the existing equivalence $\tilde{\mathcal{F}}_{F_0} \simeq \mathcal{L}$ equivariant, and sharpen the reference-transfer ambiguity to an exact unique-sign statement. The task should end by packaging the resulting one-fibre diagram and recording, without constructing them, the additional dependencies required for Task 26: a nontrivial base, a varying family of metric-oriented fibres, topology/local triviality, and compatible transition data. Task 26 can then begin a fresh decomposition of the passage from one certified frame fibre to a genuinely varying frame-family problem. No manifold-level frame bundle, principal bundle, spin structure, characteristic class, or physical spin degree of freedom is claimed in Part XXIV.

Keywords: intrinsic frames; oriented orthonormal frame; frame torsor; visible rotation group; lifted frames; double cover; reference independence; kernel sign; $SU(2)$; $SO(3)$; $Spin(3)$; Lean 4; formal verification; Conservation of Difficulty.

Contents

1	From a covering group to the object it acts on	4
2	Formal provenance, source boundary, and build status	4
3	Package A: the frozen metric-oriented carrier	5
4	Package B: reconstructing the frame carrier without the group	6
5	Packages C–E: the visible frame action closes as a free transitive action	6
5.1	Freeness	7
5.2	Transitivity	7
6	Package F: a reference frame is only a coordinate choice	7
7	Packages G–H: constructing the two-sheeted lifted frame carrier	8
7.1	Exactly two lifts	8
7.2	Changing the reference frame upstairs	9
8	Package I: the internal action on ordinary frames	9
9	The post-run review boundary: ordinary frames versus lifted frames	9
9.1	A natural paper-level candidate	10
10	Package J: reconvergence with the local data of Part XXIII	10
11	What Task XXIV proves, and what it deliberately does not	11
12	Difficulties, failed proof routes, and genuine open obligations	12
12.1	Hard point 1: transitivity without defining frames as the group	12
12.2	Hard point 2: reference independence is asymmetric downstairs and upstairs	12
12.3	Hard point 3: the action-level interpretation stops one theorem too early	12
12.4	Formal proof-engineering failures	12
12.5	Open obligations intentionally not pursued	13

13	Dependency DAG and the location of the remaining difficulty	13
14	Late conventional comparison	13
15	Conservation-of-Difficulty ledger	15
16	Task XXV: focused hardening and preparation of the next frontier	15
17	Task XXVI: the next decomposition target, not yet the next theorem	17
18	Conclusion	17
A	Principal Task-24 endpoint inventory	18
B	Source and proof-boundary audit summary	18

1 From a covering group to the object it acts on

Part XXIII closed a long local-to-global loop. The project no longer had merely a spin-like sign defect or a collection of local representatives: it had a certified two-sheeted covering map

$$p : \mathcal{L} \longrightarrow G_{\text{vis}}, \quad \ker p = \{\pm 1\}, \quad (1)$$

with $\mathcal{L}$ identified group-theoretically with the unit-quaternion core and G_{vis} with the visible rotation group [6, 7]. Task 23 had additionally reconstructed the cover from local sections and sign-valued transitions before applying standard covering terminology.

That success makes the next dependency boundary more precise. A covering group is a statement about groups and a homomorphism. A spin structure, by contrast, is a lift of a frame-level geometric object over a base. Jumping directly from (1) to a standard spin-structure definition would therefore skip the object that is supposed to be lifted. Task 24 was designed to isolate exactly one intermediate layer: the oriented orthonormal frames of the already reconstructed three-dimensional spatial carrier.

Methodological principle 1 (Object first, coordinates later). A chosen basis, chart family, reference frame, or already identified group may be used to prove properties of an intrinsic carrier, but it must not be inserted into the definition merely because it supplies a convenient proof route. The object is reconstructed first; its coordinate and group presentations are proved afterwards.

This principle is reflected directly in the source order. The frame carrier is defined before the visible action is used to prove transitivity, and before any reference frame appears. Only after the carrier exists and its group action is proved free and transitive does a reference frame induce an equivalence with G_{vis} .

The resulting task is a one-fibre problem. It deliberately does *not* introduce a manifold, tangent bundle, vector bundle, principal bundle, or varying family of fibres. This limitation is not cosmetic. It makes it possible to distinguish what has been reconstructed at the frame level from what a future bundle-level task would still need.

2 Formal provenance, source boundary, and build status

The supplied Task-24 archive has SHA-256

a3ee7966b909b3540ff96d1a3b26e62a985160270ead380b2890736db65673f0.

The toolchain is Lean 4.28.0, and the resolved Mathlib revision is

8f9d9cff6bd728b17a24e163c9402775d9e6a365.

The Task-24 source consists of eight modules and 1350 Lean lines [2]. The import chain is

`Task23.Task23` $\rightarrow$ `SafeBase` $\rightarrow$ `IntrinsicFrames` $\rightarrow$ `VisibleFrameAction` $\rightarrow$ `FrameTorsor`
 $\rightarrow$ `ReferenceIndependence` $\rightarrow$ `LiftedFrames` $\rightarrow$ `Comparison` $\rightarrow$ `Task24`. (2)

A direct directory comparison with the supplied Task-23 archive shows that the complete `NullSectorTask23` source is bitwise unchanged. The Task-24 audit reports that no file outside the new Task-24 directory, the cumulative README, and the audit itself was modified.

Check	Result	Detail
New Task-24 source	8 modules	1350 Lean lines

Check	Result	Detail
Focused build	PASS	<code>lakebuildRequestProject.NullSectorTask24.Task24</code> , 8213 jobs
Whole-project build	PASS	<code>lake build</code> , 8284 jobs
Task-24 warnings	0	114 project warnings are inherited from earlier unchanged layers
Principal endpoints	38	exposed in <code>Task24.lean</code>
Axiom baseline	exact	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code> on every audited endpoint
<code>sorry/admit</code>	0/0	no proof holes
custom axiom / proof escape	0	no <code>implemented_by</code> , <code>native_decide</code> , <code>bv_decide</code> , <code>unsafe</code> , or <code>partial</code>

Proof boundary 1 (Verification boundary). *The supplied source tree, Task-24 audit, endpoint module, cumulative README, and exact source differences were inspected for this paper. The reported 8,000-plus-job builds and axiom outputs are taken from the delivered Aristotle artifact; they were not independently rerun in the paper-generation environment. The archive hash, toolchain, Mathlib revision, source counts, and Task-23 bitwise preservation were independently checked.*

The 38 public endpoints exceed the aspirational 20–25 endpoint count in the Task-24 prompt. This is not evidence that the mathematical search broadened. The source remains within eight modules and the extra endpoints are predominantly extensionality, basis, action-law, quotient, comparison, and negative-control statements. The three intended hard targets remain identifiable as: the intrinsic frame carrier plus free/transitive visible action; reference-frame independence; and the two-sheeted lifted-frame construction.

3 Package A: the frozen metric-oriented carrier

Task 24 begins with inherited data rather than with a conventional frame group. The carrier is

$$E := \text{Vec3}, \quad (3)$$

with inherited real vector-space structure and positive-definite form h_3 . The visible carrier is packaged as G_{vis} , and the internal carrier as $\mathcal{L}$. The visible action on E , written here as

$$g \triangleright x, \quad (4)$$

is the inherited spatial action from Task 21.

The one genuinely new ingredient in Package A is a minimal orientation datum

$$[p, q, r] := h_3(p \times q, r), \quad (5)$$

where the cross-product-like operation is inherited from the previous reconstruction. The source then proves that the determinant of the matrix with columns p, q, r is exactly $[p, q, r]$, and derives preservation of this triple product under the visible action from the inherited determinant-one theorem.

This is an important dependency choice. The task does not import an abstract orientation package and does not define a frame by first identifying E with a standard oriented Euclidean space. It uses the minimum orientation information already available in the project.

Status 1 (Package A). The metric carrier, visible action, and determinant theorem are inherited. The triple product is new packaging of inherited spatial algebra, and orientation preservation is derived. No frame carrier has yet been defined, and no reference frame has been chosen.

4 Package B: reconstructing the frame carrier without the group

An ordered triple is represented by a function $f : \text{Fin}(3) \rightarrow E$. Task 24 defines

$$\text{ON}(f) : \iff \forall i, j, \quad h_3(f_i, f_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad (6)$$

$$\text{Pos}(f) : \iff 0 < [f_0, f_1, f_2]. \quad (7)$$

The intrinsic oriented-frame carrier is therefore

$$\mathcal{F}^+(E) := \{f : \text{Fin}(3) \rightarrow E \mid \text{ON}(f) \wedge \text{Pos}(f)\}. \quad (8)$$

In Lean this is `FramePlus`. By inspection of the definition, neither G_{vis} nor a reference frame occurs in (8).

A fixed inherited coordinate triple `be` is used only to prove nonemptiness. It is packaged as `coordFrame`; it does not enter the definition of `FramePlus`. The source then proves a complete expansion formula

$$x = \sum_{i=0}^2 h_3(F_i, x) F_i, \quad (9)$$

for every $F \in \mathcal{F}^+(E)$, from which spanning and linear independence follow. Every frame therefore yields a basis `frameBasis`.

The matrix M_F whose columns are the frame vectors satisfies

$$M_F^T M_F = I, \quad \det M_F = 1, \quad [F_0, F_1, F_2] = 1. \quad (10)$$

These statements are not used to redefine the frame; they are consequences of the intrinsic orthonormality and positivity predicates.

Methodological principle 2 (Proof witness versus constituent). The coordinate frame proves that $\mathcal{F}^+(E)$ is nonempty. It is a witness for one theorem about $\mathcal{F}^+(E)$, not a constituent of $\mathcal{F}^+(E)$. The later coordinate equivalence with G_{vis} must therefore be read as a theorem about an already existing carrier, not as its definition.

5 Packages C–E: the visible frame action closes as a free transitive action

The visible action is extended componentwise:

$$(g \cdot F)_i := g \triangleright F_i. \quad (11)$$

Preservation of h_3 gives orthonormality of $g \cdot F$; preservation of the triple product gives positive orientation. The inherited action laws on E therefore package into a genuine multiplicative action of G_{vis} on $\mathcal{F}^+(E)$.

5.1 Freeness

The freeness proof follows the preferred intrinsic route. If $g \cdot F = F$, then g fixes all three vectors of the frame. Because the frame spans E , linearity implies that g fixes every vector of E . Faithfulness of the inherited visible representation then gives $g = 1$. Formally,

$$g \cdot F = F \implies g = 1. \quad (12)$$

The corresponding stabilizer is therefore trivial.

This proof does not import freeness from a standard $SO(3)$ -torsor theorem. Its dependencies are the reconstructed frame basis, linearity, and faithfulness of the inherited visible action.

5.2 Transitivity

Transitivity is more expensive. Given $F, G \in \mathcal{F}^+(E)$, Task 24 constructs the explicit matrix

$$A_{F \rightarrow G} := M_G M_F^\top. \quad (13)$$

The source proves

$$A_{F \rightarrow G} A_{F \rightarrow G}^\top = I, \quad \det A_{F \rightarrow G} = 1, \quad A_{F \rightarrow G} M_F = M_G. \quad (14)$$

Thus $A_{F \rightarrow G}$ is an orientation-preserving orthogonal transformation carrying the first frame to the second. At precisely this point the proof uses the already certified surjectivity of the visible carrier onto the standard special-orthogonal matrix model. The result is

$$\forall F, G \in \mathcal{F}^+(E), \quad \exists g \in G_{\text{vis}} : g \cdot F = G. \quad (15)$$

Combining (12) and (15) gives the sharper theorem

$$\forall F, G \in \mathcal{F}^+(E), \quad \exists! g \in G_{\text{vis}} : g \cdot F = G. \quad (16)$$

The unique element is packaged as `frameHom F G`.

Methodological caveat 1 (Definitions and proof routes remain distinct). *The definition of $\mathcal{F}^+(E)$ is independent of G_{vis} . The transitivity proof is not: its final membership step uses the inherited surjectivity theorem identifying the visible group with the orientation-preserving orthogonal matrix model. This is a proof-route dependency, not a definitional dependency of the frame carrier.*

In conventional terminology, (12)–(16) say that $\mathcal{F}^+(E)$ is a nonempty free transitive G_{vis} -set. Task 24 deliberately does not spend proof budget adapting a general torsor API; the structural content is nevertheless proved directly.

6 Package F: a reference frame is only a coordinate choice

Only after free transitivity has been established does the source choose a reference frame $F_0 \in \mathcal{F}^+(E)$. It defines

$$c_{F_0} : G_{\text{vis}} \rightarrow \mathcal{F}^+(E), \quad c_{F_0}(g) = g \cdot F_0. \quad (17)$$

Freeness gives injectivity, transitivity gives surjectivity, and therefore

$$c_{F_0} : G_{\text{vis}} \simeq \mathcal{F}^+(E) \quad (18)$$

is an equivalence of underlying sets.

For a second reference frame F_1 , let

$$h_{01} := \text{frameHom}(F_0, F_1), \quad h_{01} \cdot F_0 = F_1. \quad (19)$$

Then Task 24 proves the exact change-of-reference formula

$$c_{F_1}(g) = c_{F_0}(gh_{01}). \quad (20)$$

Thus different choices of reference frame produce coordinate presentations related by a uniquely determined right translation.

Status 2 (Hard target B). The reference frame does not define the frame carrier. It selects an origin in a free transitive G_{vis} -set and thereby turns that set into a coordinate copy of the group. Changing the origin changes the coordinate map by one uniquely determined group element.

The source explicitly stops before a topological strengthening. No topology is defined on $\mathcal{F}^+(E)$, so (18) is not a homeomorphism theorem. Constructing a topology merely to upgrade the equivalence was correctly judged to be a separate development rather than a necessary part of Task 24.

7 Packages G–H: constructing the two-sheeted lifted frame carrier

The next step uses the already certified group projection $p : \mathcal{L} \rightarrow G_{\text{vis}}$. For a temporary reference frame F_0 , Task 24 defines

$$\tilde{\mathcal{F}}_{F_0} := \{(u, F) \in \mathcal{L} \times \mathcal{F}^+(E) \mid p(u) \cdot F_0 = F\}. \quad (21)$$

This is deliberately not defined to be $\mathcal{L}$, even though the source later proves that the internal component is a complete invariant once F_0 has been fixed.

The projection to ordinary frames is

$$\pi_{F_0} : \tilde{\mathcal{F}}_{F_0} \rightarrow \mathcal{F}^+(E), \quad \pi_{F_0}(u, F) = F. \quad (22)$$

Surjectivity follows from surjectivity of p and the unique visible frame transformation.

7.1 Exactly two lifts

Let

$$K_{\pm} := \ker p = \{\pm 1\}. \quad (23)$$

The source defines the vertical sign action by

$$\varepsilon \cdot (u, F) := (\varepsilon u, F), \quad \varepsilon \in K_{\pm}. \quad (24)$$

It proves:

1. the sign action is free;
2. $x, y \in \tilde{\mathcal{F}}_{F_0}$ have the same ordinary frame iff there exists a unique $\varepsilon \in K_{\pm}$ with $y = \varepsilon \cdot x$;
3. every fibre of π_{F_0} has exactly two elements;
4. the quotient of $\tilde{\mathcal{F}}_{F_0}$ by the same-frame/sign relation is equivalent to $\mathcal{F}^+(E)$.

In symbols,

$$\tilde{\mathcal{F}}_{F_0}/K_{\pm} \simeq \mathcal{F}^+(E) \quad (25)$$

at the set/equivalence level.

A further equivalence is immediate from the compatibility condition:

$$\text{elt} : \tilde{\mathcal{F}}_{F_0} \simeq \mathcal{L}, \quad (26)$$

with inverse $u \mapsto (u, p(u) \cdot F_0)$. This result will become important in the review boundary below.

7.2 Changing the reference frame upstairs

Downstairs, the reference change $h_{01} \in G_{\text{vis}}$ is unique. Upstairs, Task 24 chooses an internal element $u_0 \in \mathcal{L}$ satisfying

$$p(u_0) = h_{01}. \quad (27)$$

Given this choice, it constructs an explicit equivalence

$$T_{u_0} : \tilde{\mathcal{F}}_{F_0} \simeq \tilde{\mathcal{F}}_{F_1}, \quad T_{u_0}(u, F) = (uu_0^{-1}, F). \quad (28)$$

The transfer preserves the ordinary frame and commutes with the vertical kernel-sign action.

If u'_0 is another lift of the same visible reference change, the source proves that there exists a kernel sign ε such that

$$T_{u'_0}(x) = \varepsilon \cdot T_{u_0}(x) \quad \text{for all } x. \quad (29)$$

This is the precise form of reference independence proved by Task 24.

Methodological caveat 2 (What “canonical up to sign” should mean). *The ordinary reference change h_{01} is unique. Its internal lift is not: there are two possibilities differing by $K_{\pm}$. Accordingly, the lifted comparison is not a singled-out canonical map unless an internal lift is chosen. Task 24 proves existence of a sign relating any two such comparison maps. A stronger uniqueness-of-that-sign theorem is natural but was not exposed as a Task-24 endpoint.*

This is not a defect of the construction. It is the same two-sheeted information that the project has been tracking since the earlier sign obstruction, now appearing as the ambiguity in lifting a reference change.

8 Package I: the internal action on ordinary frames

Task 24 next defines the direct internal action on the *ordinary* frame carrier through the certified projection:

$$u \star F := p(u) \cdot F. \quad (30)$$

Surjectivity of p and transitivity of the visible action imply that this action is transitive. It cannot be free, because every element of $K_{\pm}$ projects to the identity. Task 24 proves the exact stabilizer theorem

$$\text{Stab}_{\mathcal{L}}(F) = K_{\pm} \quad \text{for every } F \in \mathcal{F}^+(E), \quad (31)$$

and the exact fibre relation

$$u \star F = v \star F \iff \exists! \varepsilon \in K_{\pm} : v = \varepsilon u. \quad (32)$$

The explicit negative control `lact_not_free` exhibits the nontrivial kernel sign fixing every ordinary frame.

These are correct and important theorems. They say that the internal group acts on ordinary frames through its visible quotient, so the kernel is necessarily invisible downstairs.

9 The post-run review boundary: ordinary frames versus lifted frames

The Task-24 audit goes one interpretive step beyond the theorem (31). Its action table leaves the row “ $\mathcal{L}$ acts on $\tilde{\mathcal{F}}_{F_0}$ ” without a formal action, and its prose describes (26) as the strongest correct comparison because the internal action on ordinary frames has stabilizer $K_{\pm}$. This conclusion is too weak as an interpretation of the *lifted* carrier.

The distinction is elementary but essential:

- Task 24 *does* define and prove non-freeness of the $\mathcal{L}$ -action on $\mathcal{F}^+(E)$.
- Task 24 *does not* define a $\mathcal{L}$ -action on $\tilde{\mathcal{F}}_{F_0}$.
- Therefore `lact_not_free` says nothing about freeness of a possible action on $\tilde{\mathcal{F}}_{F_0}$.

No Lean theorem is false here. The issue is a missing formalization and an over-restrictive audit interpretation.

9.1 A natural paper-level candidate

The completed definitions suggest the following action candidate:

$$a \blacktriangleright (u, F) := (au, p(a) \cdot F). \quad (33)$$

If $(u, F) \in \tilde{\mathcal{F}}_{F_0}$, then $F = p(u) \cdot F_0$, and hence

$$p(au) \cdot F_0 = p(a)p(u) \cdot F_0 \quad (34)$$

$$= p(a) \cdot F, \quad (35)$$

so the right-hand side again satisfies the defining compatibility condition (21). The action laws are inherited from group multiplication and the visible action.

Moreover, the same algebra strongly indicates freeness. If

$$a \blacktriangleright (u, F) = (u, F), \quad (36)$$

then equality of internal components gives $au = u$, hence $a = 1$. For transitivity, given $x = (u, F)$ and $y = (v, G)$, the candidate

$$a := vu^{-1} \quad (37)$$

sends the internal component of x to v , and the frame component follows from the compatibility conditions.

Finally,

$$\pi_{F_0}(a \blacktriangleright x) = p(a) \cdot \pi_{F_0}(x), \quad (38)$$

and restricting (33) to $a \in K_{\pm}$ should recover the already formalized vertical sign action (24).

Proof boundary 2 (Review result, not Task-24 theorem). *Equations (33)–(38) and the free/transitive argument above are paper-level diagnostics derived from the Task-24 definitions. They are not declarations in the supplied Task-24 source. Because an Aristotle task is immutable once completed, these statements belong to a new Task 25 hardening run.*

This review point is mathematically productive. The non-free action on ordinary frames and the expected free action on lifted frames are not in tension. The extra sign sheet is exactly the information that removes the kernel stabilizer upstairs.

10 Package J: reconvergence with the local data of Part XXIII

Task 24 keeps the comparison with the Task-23 atlas intentionally small. It does not rebuild the four-chart cover. Instead it packages the inherited local representative as a lifted frame after a reference frame is chosen and proves one overlap theorem.

If $s_i(g)$ and $s_j(g)$ are the Task-23 local lifts of the same visible point, with transition sign $\tau_{ij}(g)$, the corresponding Task-24 lifted frames satisfy

$$\tilde{s}_j(g) = \kappa(\tau_{ij}(g)) \cdot \tilde{s}_i(g), \quad (39)$$

where κ converts the Task-23 sign into the packaged kernel element. In particular both local choices project to the same ordinary frame.

This theorem closes an important reconvergence. The sign that glued local sections of the group cover in Part XXIII is the same vertical sign that distinguishes the two lifted representatives of a frame in Part XXIV. The four-chart atlas remains a proof route rather than a constituent of the frame definition.

11 What Task XXIV proves, and what it deliberately does not

Statement	Status	Principal source
Intrinsic oriented orthonormal frame carrier $\mathcal{F}^+(E)$ constructed	PROVED	<code>IntrinsicFrames</code>
Definition of $\mathcal{F}^+(E)$ independent of G_{vis}	PROVED by definition	<code>FramePlus</code>
Definition of $\mathcal{F}^+(E)$ independent of reference frame	PROVED by definition	<code>FramePlus</code>
Every frame is a basis of E	PROVED	<code>frameBasis</code>
G_{vis} acts on $\mathcal{F}^+(E)$	PROVED	<code>VisibleFrameAction</code>
Visible action is free	PROVED	<code>frameAction_free</code>
Visible action is transitive	PROVED	<code>frameAction_transitive</code>
Unique visible transformation between two frames	PROVED	<code>existsUnique_gvis_map_frame</code>
Reference frame gives $G_{\text{vis}} \simeq \mathcal{F}^+(E)$	PROVED, set level	<code>coordEquiv</code>
Change of reference classified by right translation	PROVED	<code>coord_change</code>
Lifted-frame carrier constructed	PROVED	<code>LiftedFrame</code>
Exactly two lifted representatives per frame	PROVED	<code>two_lifts</code>
Same-frame lifts differ by a unique sign	PROVED	<code>liftedFrame_eq_iff_sign</code>
Vertical sign action free	PROVED	<code>signSmul_free</code>
Lifted quotient recovers ordinary frames	PROVED, set level	<code>liftedQuotientEquiv</code>
Fixed-reference lifted carrier equivalent to $\mathcal{L}$	PROVED, set level	<code>liftedEltEquiv</code>
Lifted carriers for two references explicitly equivalent	PROVED	<code>liftedTransfer</code>
Reference-transfer maps differ by a kernel sign	PROVED, existence of sign	<code>liftedTransfer_choice</code>
$\mathcal{L}$ acts transitively on ordinary frames	PROVED	<code>lact_transitive</code>
Stabilizer on ordinary frames is exactly $K_{\pm}$	PROVED	<code>lact_stabilizer</code>
$\mathcal{L}$ -action on ordinary frames is not free	PROVED	<code>lact_not_free</code>
$\mathcal{L}$ -action on lifted frames defined	NOT IN TASK 24	Task-25 hardening
Free/transitive $\mathcal{L}$ -action on lifted frames	NOT IN TASK 24	Task-25 hardening
Topology on $\mathcal{F}^+(E)$	NOT DEVELOPED	open boundary
Topological torsor/homeomorphism statement	NOT DEVELOPED	open boundary
Frame bundle over a nontrivial base	NO	outside scope
Spin structure	NO	outside scope
Physical spin degree of freedom	NO	outside scope

This table is the theorem boundary for Part XXIV. In particular, the two paper-level candidate rows are intentionally marked as absent from Task 24 even though their algebra appears straightforward.

12 Difficulties, failed proof routes, and genuine open obligations

The task stayed within budget because it did not confuse the number of lemmas with the number of hard mathematical branches. Most of the 38 endpoint declarations package consequences around three difficult transitions: proving free/transitive visible action, controlling reference independence, and classifying the two-sheeted lifted carrier.

12.1 Hard point 1: transitivity without defining frames as the group

Freeness is relatively direct once every frame is known to span E . Transitivity is harder because the task is forbidden to set $\mathcal{F}^+(E) := G_{\text{vis}}$. It must construct a transformation between arbitrary frames, prove that this transformation preserves the metric and orientation, and only then show that it belongs to the already reconstructed visible carrier. The proof therefore legitimately uses the inherited surjectivity theorem $G_{\text{vis}} \rightarrow SO(3)$ only at the final membership step.

This is precisely the desired balance between intrinsic reconstruction and reuse of already certified structure. Reproving the full visible-group identification would have duplicated Task 21; defining frames as group elements would have made Task 24 tautological.

12.2 Hard point 2: reference independence is asymmetric downstairs and upstairs

Downstairs, free transitivity makes the change-of-reference element unique. Upstairs, lifting that element through a two-sheeted group projection necessarily introduces two choices. Task 24 controls the ambiguity, but its theorem `liftedTransfer_choice` asserts existence of a kernel sign relating two comparison maps rather than exposing a unique-sign classification as a public endpoint.

This distinction should remain visible. The residual sign is not a nuisance introduced by coordinates; it is the information retained by the double cover.

12.3 Hard point 3: the action-level interpretation stops one theorem too early

Package I correctly finds the kernel stabilizer on ordinary frames. The source simultaneously proves $\tilde{\mathcal{F}}_{F_0} \simeq \mathcal{L}$, but does not combine these ingredients into an action theorem upstairs. The post-run review therefore identifies a narrow missing closure statement rather than a failed construction. This is the main item exported to Task 25.

12.4 Formal proof-engineering failures

The Task-24 audit records four kinds of failed proof attempts or build repairs. None changed a mathematical theorem statement.

1. `liftedTransfer_proj` was initially attempted by `rfl`. Definitional reduction expanded frame values depending on `frameHom`, which is built from `Classical.choose`; the kernel timed out. The theorem statement was retained and the proof was rewritten propositionally from the lifted-frame compatibility condition and group law.
2. The installed Mathlib revision uses `Module.Basis`, not the guessed unqualified `Basis`; the corresponding constructor/application names were repaired.
3. Definitions depending on the certified noncomputable projection required explicit `noncomputable` markers.
4. Two explicit coordinate reductions required additional matrix simplification lemmas for the third component before `norm_num` could close them.

These failures are engineering provenance, not mathematical counterexamples. The important mathematical review point is different: no theorem failed, but the final audit interpretation left the natural action on the lifted carrier unformalized.

12.5 Open obligations intentionally not pursued

Task 24 also records several genuine scope boundaries:

- no topology on $\mathcal{F}^+(E)$, hence no homeomorphism version of $G_{\text{vis}} \simeq \mathcal{F}^+(E)$;
- no general torsor typeclass packaging;
- no choice-free singled-out reference-transfer map upstairs;
- only one compatibility theorem with the Task-23 local cover;
- no varying base and therefore no frame bundle.

The first and fifth items are potentially substantial new developments. They should not be mixed into the small algebraic hardening needed immediately after Task 24.

13 Dependency DAG and the location of the remaining difficulty

The DAG shows where the Conservation of Difficulty moved. Task 23 had difficulty in gluing local representatives into a global double cover. Task 24 no longer struggles to construct frames or to make the visible group act on them: those parts close. The surviving issue is the exact upstairs action and its reference-change coherence. Only after that one-fibre structure is closed does it make sense to ask how such fibres could vary over another space.

14 Late conventional comparison

Only after the intrinsic reconstruction is fixed is standard terminology useful. A nonempty set with a free transitive action of a group is conventionally a torsor or principal homogeneous space. Thus the Task-24 theorems establish the algebraic content of a G_{vis} -torsor on $\mathcal{F}^+(E)$, even though no general torsor API is installed in the Lean layer [4].

Likewise, because previous tasks identify the core group with the unit quaternions and hence classically with $SU(2) \cong Spin(3)$, while $G_{\text{vis}} \cong SO(3)$, the diagram

$$\mathcal{L} \longrightarrow G_{\text{vis}} \quad (40)$$

is the group-level substrate of the classical $Spin(3) \rightarrow SO(3)$ double cover [5, 3]. Part XXIV adds a single oriented-frame fibre and its two-sheeted lifted version. This is closer to the language used in spin geometry, but it is still not a spin structure.

A spin structure on a manifold requires, in one standard formulation, an oriented orthonormal frame bundle over a base together with a compatible lift through the spin double cover [5]. Task 24 has no nontrivial base, no tangent bundle, no varying fibre family, and no bundle projection over a manifold. The statement

$$\tilde{\mathcal{F}}_{F_0}/K_{\pm} \simeq \mathcal{F}^+(E) \quad (41)$$

is therefore a one-fibre algebraic result. It must not be promoted into a manifold-level spin claim.

Methodological caveat 3 (No physical inference). *Nothing in Task 24 constructs a Hilbert space, spin observable, fermionic statistics, quantum state, Hamiltonian, field equation, mass, or charge. The word “spin” enters only in the late mathematical comparison of an already reconstructed group double cover with the classical $Spin(3)$ model.*

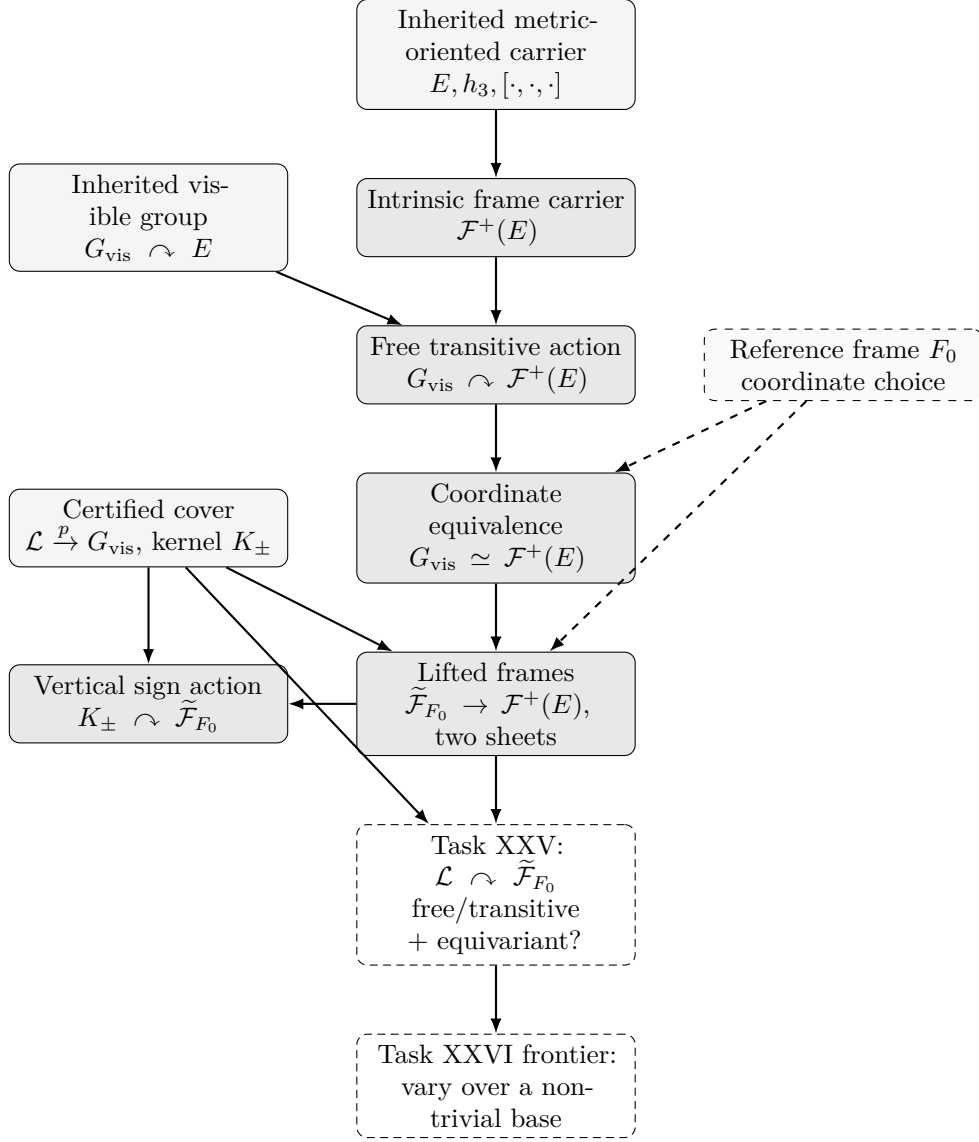


Figure 1: Definition/dependency structure of Task 24 and the narrow hardening boundary. Dashed boxes/arrows denote coordinate choices or still-unproved next-layer statements, not constituents of the intrinsic frame definition.

15 Conservation-of-Difficulty ledger

Question	Part-XXIV status
Must the oriented frame carrier be defined from G_{vis} ?	No. It is defined intrinsically from E, h_3 , and orientation.
Does a coordinate/reference frame belong to the definition?	No. One coordinate frame proves nonemptiness; later reference frames only coordinatize.
Does G_{vis} act on the intrinsic frames?	Yes.
Is the visible action free?	Yes.
Is it transitive?	Yes. The hard membership step uses inherited $SO(3)$ -surjectivity.
Is there a unique visible map between any two frames?	Yes.
Does choosing F_0 identify the frame carrier with G_{vis} ?	Yes, as an underlying-set equivalence, but only after the intrinsic object is already defined.
Is the change of reference downstairs unique?	Yes.
Can each ordinary frame be lifted through $\mathcal{L} \rightarrow G_{\text{vis}}$?	Yes, in exactly two ways.
What distinguishes the two lifts?	A unique element of $K_{\pm} = \{\pm 1\}$.
Does the vertical kernel-sign action act freely?	Yes.
Does the quotient recover ordinary frames?	Yes, at set/equivalence level.
Are lifted-frame constructions from two reference frames equivalent?	Yes, after choosing an internal lift of the visible reference change.
Is that comparison map uniquely determined?	Not as a singled-out map; two internal choices differ by a kernel sign.
Does $\mathcal{L}$ act freely on ordinary frames?	No; stabilizer exactly $K_{\pm}$.
Was a $\mathcal{L}$ -action on lifted frames formalized?	No. This is the principal review/hardening boundary.
Does the existing source strongly suggest such an action is free and transitive?	Yes, at paper-diagnostic level; must be proved in Task 25.
Is a topology on $\mathcal{F}^+(E)$ available?	No.
Is a varying frame bundle over a base available?	No.
Where did the difficulty move?	From constructing the frame fibre to closing the upstairs action/coherence, then to the still-absent varying-base layer.

The ledger illustrates the same structural phenomenon encountered repeatedly in this project: successful reconstruction does not remove difficulty; it relocates it to the first datum that is not already forced. Here the frame carrier and visible torsor close. The remaining algebraic gap is small. The later geometric gap – variation over a base – is much larger and should not be conflated with that small hardening step.

16 Task XXV: focused hardening and preparation of the next frontier

Task 25 should be a new immutable task. Its purpose is not to repeat Task 24 or to begin a bundle theory. It should close the exact action/coherence statements that Task 24 left one level short, and then package the completed one-fibre structure in a form that Task 26 can decompose further.

A balanced Task 25 should have three substantive hardening targets and one lightweight frontier package.

XXV-A: internal action on the lifted frame carrier

Define, for fixed F_0 ,

$$a \blacktriangleright (u, F) := (au, p(a) \cdot F). \quad (42)$$

Prove only the following core facts:

1. well-definedness and action laws;
2. freeness;
3. transitivity;
4. projection equivariance $\pi(a \blacktriangleright x) = p(a) \cdot \pi(x)$;
5. restriction to $K_{\pm}$ agrees exactly with the existing vertical `signSmul` action.

These are expected to be short group-algebra proofs from the existing compatibility predicate. They should not be expanded into general torsor theory.

XXV-B: make the existing equivalence equivariant

The already proved

$$\text{liftedEltEquiv} : \tilde{\mathcal{F}}_{F_0} \simeq \mathcal{L} \quad (43)$$

should be shown to intertwine the new lifted-frame action with left multiplication of $\mathcal{L}$ on itself. This gives a sharp formal statement that the lifted-frame carrier is a free transitive $\mathcal{L}$ -object after a reference has been chosen, while the ordinary frame carrier is its quotient by the kernel sign.

XXV-C: harden reference transfer

For F_0, F_1 , prove that `liftedTransfer` is equivariant for the new $\mathcal{L}$ -action. Then sharpen the current choice theorem so that the residual kernel sign relating two chosen lifts of the same reference change is classified with the strongest inexpensive uniqueness statement. One suitable endpoint is that the family of comparison equivalences is acted on freely and transitively by $K_{\pm}$; if this requires unnecessary infrastructure, an explicit $\exists! \varepsilon$ theorem for two comparison choices is sufficient.

XXV-D: freeze the one-fibre interface for Task XXVI

After A–C are green, Task 25 should expose one compact diagram as a certified endpoint:

$$\begin{array}{ccc} \tilde{\mathcal{F}}_{F_0} & \xrightarrow{\pi} & \mathcal{F}^+(E) \\ \uparrow \text{dashed} & & \uparrow \text{dashed} \\ \mathcal{L} & \xrightarrow{p} & G_{\text{vis}} \end{array} \quad (44)$$

with free transitive actions upstairs/downstairs, equivariance, and the vertical kernel-sign quotient explicitly recorded.

The preparation for Task 26 should then be a dependency ledger, not a new construction. It should state that the following data remain absent:

- a nontrivial base space B ;
- a family of three-dimensional metric-oriented fibres E_b varying with $b \in B$;
- a topology on the frame and lifted-frame total spaces compatible with that variation;
- local trivializations over open subsets of B ;
- overlap/transition data between those trivializations.

Task 25 must stop before supplying these data.

Status 3 (Task-25 stop condition). When the lifted-frame action is formally free/transitive, the projection is equivariant, reference transfers are equivariant and sign-classified, and the one-fibre dependency interface is packaged, Task 25 is complete. Topology on $\mathcal{F}^+(E)$, a nontrivial base, and bundle construction remain for a later task.

17 Task XXVI: the next decomposition target, not yet the next theorem

Task 26 should begin from the hardened one-fibre interface (44) and ask a different kind of question:

What additional dependencies are minimally required to replace one oriented frame fibre by a family of such fibres over a nontrivial base, and which of those dependencies are already present, reconstructible, independent, or genuinely new?

This is deliberately not yet the instruction “construct a spin structure.” The first purpose of Task 26 should be decomposition: identify the base, fibre-family, topology, local-triviality, and transition-data branches and determine their logical arrows. Only after those nodes exist can a later task test whether the hardened double-cover data provide a compatible lift.

The distinction matters for both mathematics and proof budget. Task 25 is expected to close a small algebraic omission. Task 26 introduces the possibility of a genuinely new geometric layer. Combining them would risk masking a simple hardening problem inside a much broader search.

18 Conclusion

Part XXIV marks the first point in the project where an oriented frame object exists independently of the group that later coordinates it. Starting only from the inherited three-dimensional metric carrier and orientation datum, Task 24 constructs $\mathcal{F}^+(E)$, proves every frame is a basis, and establishes a free transitive G_{vis} -action. A reference frame is introduced only after this closure and is proved to supply a coordinate equivalence rather than an intrinsic choice.

The certified double cover then produces a two-sheeted lifted-frame carrier. Every ordinary frame has exactly two lifts, the two are related by a unique kernel sign, the vertical sign action is free, and the quotient recovers the ordinary frame carrier. Changing the reference frame upstairs is possible by an explicit equivalence, but the choice of internal lift of the visible reference change leaves exactly the expected sign ambiguity. The Task-23 local transition signs reconverge with this vertical ambiguity through a direct compatibility theorem.

The formal theorem package is green. Its main review issue is not a false theorem but a missing final action theorem. Task 24 proves that $\mathcal{L}$ acts transitively and non-freely on ordinary frames with stabilizer $K_{\pm}$; it does not define the natural $\mathcal{L}$ -action on the lifted-frame carrier. The existing compatibility definition and equivalence $\tilde{\mathcal{F}}_{F_0} \simeq \mathcal{L}$ strongly suggest that this upstairs action should instead be free and transitive. Because the task is immutable, that strengthening belongs to Task 25.

The resulting frontier is now exceptionally clean. Task 25 should close the one-fibre algebra and reference coherence. Task 26 can then start from a fully hardened frame/lift diagram and decompose the genuinely new problem of variation over a nontrivial base. The project has therefore not yet reached a spin structure, but it has isolated the exact mathematical object that must be made to vary before such a question can be posed without importing the answer.

A Principal Task-24 endpoint inventory

The endpoint module exposes 38 audited declarations. They are grouped below by conceptual role rather than treated as 38 independent research results.

Package	Principal declarations
Intrinsic frames	<code>task24_framePlus_nonempty</code> , <code>task24_frame_ext</code> , <code>task24_frame_span</code> , <code>task24_frame_linearIndependent</code> , <code>task24_frame_expand</code> , <code>task24_frameBasis</code>
Visible action	<code>task24_frames_orthonormal_preserved</code> , <code>task24_frames_orientation_preserved</code> , <code>task24_frame_action_one</code> , <code>task24_frame_action_mul</code>
Free/transitive closure	<code>task24_frameAction_free</code> , <code>task24_frameAction_transitive</code> , <code>task24_existsUnique_gvis_map_frame</code> , <code>task24_framePlus_free_transitive</code> , <code>task24_stabilizer_frame_eq_bot</code>
Reference coordinates	<code>task24_coordEquiv</code> , <code>task24_coord_bijective</code> , <code>task24_change_of_reference</code> , <code>task24_change_of_reference_equiv</code> , <code>task24_reference_independence</code> , <code>task24_reference_not_intrinsic</code>
Lifted frames	<code>task24_liftedFrameProj_surjective</code> , <code>task24_two_lifts</code> , <code>task24_lifted_fibre_ncard</code> , <code>task24_lifts_differ_by_unique_sign</code> , <code>task24_sign_action_free</code> , <code>task24_liftedQuotientEquiv</code>
Lifted reference change	<code>task24_liftedTransfer</code> , <code>task24_liftedTransfer_proj</code> , <code>task24_liftedTransfer_sign</code> , <code>task24_liftedTransfer_choice</code>
Internal action downstairs	<code>task24_lact_transitive</code> , <code>task24_lact_stabilizer</code> , <code>task24_lact_eq_iff</code> , <code>task24_lact_not_free</code>
Comparison/controls	<code>task24_local_cover_compatibility</code> , <code>task24_no_gvis_fixed_frame</code> , <code>task24_liftedFrameProj_not_injective</code>

Every declaration in this endpoint module reports exactly the baseline axioms `propext`, `Classical.choice`, and `Quot.sound`; none reports `sorryAx`.

B Source and proof-boundary audit summary

- New source: 8 modules, 1350 lines.
- Input: Task 23 imported unchanged; direct directory comparison is bitwise identical.
- Final focused build: 8213 jobs, PASS.
- Final whole-project build: 8284 jobs, PASS.

- Task-24 warnings: 0.
- Prohibited constructs in Task-24 source: 0 occurrences of `sorry`, `admit`, `custom axiom`, `implemented_by`, `native_decide`, `bv_decide`, `unsafe`, or `partial`.
- No theorem statement was weakened in response to a failed proof attempt.
- The principal review correction concerns interpretation of the absent lifted-frame $\mathcal{L}$ -action, not validity of a compiled theorem.

References

- [1] Aristotle (Harmonic). *Task 23: Local Decomposition and Reconstruction of the Certified Double Cover*. Run c3139bdc-c8f5-4e55-83bc-eeb637a826dc; focused build 8205 jobs; whole-project build 8276 jobs; 55 principal endpoints; archive SHA-256 c8312fe9dcd5d76b623066ffd73ba944a96a751f87aa3dc1fa363bfb5ee6097. 2026.
- [2] Aristotle (Harmonic). *Task 24: Intrinsic Oriented Frame Torsor and Lifted Frame Reconstruction*. Formal Lean archive supplied by the editor on 5 September 2026; 8 Task-24 modules, 1350 lines, 38 principal endpoints; focused build 8213 jobs; whole-project build 8284 jobs; archive SHA-256 a3ee7966b909b3540ff96d1a3b26e62a985160270ead380b2890736db65673f0. 2026.
- [3] Brian C. Hall. *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*. 2nd ed. Vol. 222. Graduate Texts in Mathematics. Springer, 2015.
- [4] Shoshichi Kobayashi and Katsumi Nomizu. *Foundations of Differential Geometry, Volume I*. Interscience Publishers, 1963.
- [5] H. Blaine Lawson and Marie-Louise Michelsohn. *Spin Geometry*. Princeton University Press, 1989.
- [6] Jan Seeck, OpenAI ChatGPT, and Aristotle (Harmonic). *Split Complex Lorentzian Dynamics – Part XXI: The Quaternionic Double-Cover Identification, Completed and Certified*. Internal project manuscript, updated 5 September 2026; merges the interrupted Task-21 reconstruction with Task-22 completion and certification. Sept. 2026.
- [7] Jan Seeck, OpenAI ChatGPT, and Aristotle (Harmonic). *Split Complex Lorentzian Dynamics – Part XXIII: Local Sections, Transition Signs, and Exact Reconstruction of the Quaternionic Double Cover*. Internal project manuscript, revised 5 September 2026. Sept. 2026.